RESEARCH ARTICLE

Does urbanization shape bacterial community composition in urban park soils? A case study in 16 representative Chinese cities based on the pyrosequencing method

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Abstract

Although the geographical distribution patterns of microbes have been studied for years, few studies have focused on urban soils. Urbanization may have detrimental effects on the soil ecosystem through pollution discharge and changes in urban climate. It is unclear whether urbanization-related factors have any effect on soil bacterial communities. Therefore we investigated geographical patterns of soil microbial communities in parks in 16 representative Chinese cities. The microbial communities in these 95 soil samples were revealed by 454-pyrosequencing. There were 574 442 effective sequences among the total of 980 019 16S rRNA gene sequences generated, showing the diversity of the microbial communities. *Proteobacteria*, *Actinobacteria*, *Acidobacteria*, *Planctomycetes*, *Chloroflexi* and *Bacteroidetes* were found to be the six dominant phyla in all samples. Canonical correspondence analysis showed that pH, followed by annual average precipitation, annual average temperature, annual average relative humidity and city sunshine hours, Mn and Mg were the factors most highly correlated with the bacterial community variance. Urbanization did have an effect on bacterial community composition of urban park soils but it contributed less to the total variance compared with geographical locations and soil properties, which explained 6.19% and 16.78% of the variance, respectively.

Introduction

The dominant role of microorganisms, especially bacteria, in terrestrial and aquatic biogeochemical cycles has prompted an increasing number of studies on the diversity and biogeography of soil bacterial communities (Fierer & Jackson, 2006; Lauber *et al.*, 2009; Chu *et al.*, 2010; Shen *et al.*, 2012). Although they have been studied for many years, the biogeographical patterns of bacterial communities at continental scales are still not as clear as those of plants and animals due to the limitation of techniques and site-specificity. Since the advent of 454-pyrosequencing in 2006 (Sogin *et al.*, 2006), high throughput sequencing technologies and advanced bioinformatics methods have been widely and efficiently used to study geographical patterns of microbial communities (Lauber *et al.*, 2009).

Although geographical distributions of microorganisms and the biotic and abiotic factors influencing them are poorly understood, attempts have been made to understand how soil microbial diversity varies across the globe and how the diversity is related to the physical, chemical and biological characteristics of ecosystems (Shen *et al.*, 2012). It is generally held that soil pH plays an important role in determining soil bacterial spatial distribution at both vertical (Shen *et al.*, 2012) and horizontal gradients, including at local (Rousk *et al.*, 2010; Osborne *et al.*, 2011), regional (Chu *et al.*, 2010; Tripathi *et al.*, 2012) and continental scales (Lauber *et al.*, 2009). Soil pH is not the only factor that can influence bacterial community composition and diversity. The types and quantities of organic carbon can also contribute much to changes in soil bacterial communities (Fierer *et al.*, 2007; Goldfarb *et al.*, 2011; Tripathi *et al.*, 2012). Other factors such as soil temperature, moisture, C/N ratio, nutrient availability, and plants are also shown to have considerable influence on soil microbial communities (Högberg *et al.*, 2007; Lauber *et al.*, 2008; Fierer *et al.*, 2012). Climate change factors, including elevated CO_2, temperature and precipitation, have been observed to affect soil microbial community composition in some specific ecosystems (Cruz-Martinez *et al.*, 2009; Rinnan *et al.*, 2009; Castro *et al.*, 2010; Kuffner *et al.*, 2012; Yergeau *et al.*, 2012). Fierer & Jackson (2006) studied the influence of site temperature and latitude on bacterial communities but found no correlation between bacterial diversity and these factors. They did find that community composition was largely independent of geographic distance. There are still not enough studies to support whether geographical distance has an effect on microbial community composition. Therefore, more studies are needed on this topic.

As described above, previous studies have focused mainly on soil properties as well as latitude and climatic factors, which are very important predictors of animal and plant distribution patterns (Gaston, 2000; Allen *et al.*, 2002; Hawkins *et al.*, 2003). In addition to environmental variables, human activities have also exerted a great influence on soil microbial communities, e.g. fertilization (Jangid *et al.*, 2008) and tillage management (Drijber *et al.*, 2000). Among human activities, the effect of urbanization has received the least attention, not only on soil microbial communities but also on ecosystems in general (Pickett & Cadenasso, 2006). It is generally believed that urbanization may have detrimental effects on soil ecosystems through pollution discharge and changes in urban climate (Civerolo *et al.*, 2007; Ash *et al.*, 2008). Despite the rapid urbanization in China, both in scale and speed (Zhu *et al.*, 2011) , little is known about soil microbial communities in urban environments and the impact of urbanization on soil ecosystems. In addition, few studies have been carried out on variation of the soil bacterial community composition in urban areas, such as parks. China offers a significant opportunity to address these issues across wide geographical and climatic scales. Furthermore, China is now and will continue experiencing drastic population shifts from rural to urban areas, with urban soil serving more and more important environmental and ecological functions (Kelly *et al.*, 1996; Mielke *et al.*, 1999). Urban park soil, which is neither sealed nor compacted, plays a vital part in urban ecosystems and thus has attracted much attention (Chen *et al.*, 2005; Li *et al.*, 2012). Although appearing to be similar to rural soil, soils in natural parks, which are seldom disturbed by human activities, receive different atmospheric deposits and are influenced by urban microclimates. It is reasonable to assume that the microbial community composition of urban park soils may be quite different from that of rural soils.

As far as we are aware, little has been done towards documenting soil bacteria communities and determining the factors driving the composition of microbial communities in urban park soils at a continental scale. To find out whether some indirect factors related to urbanization (hereafter defined as urbanization factors) are related to soil bacterial communities in a statistically meaningful way, and what are the most important soil properties, geographical factors and human/urbanization factors for predicting microbial community composition and diversity, we applied canonical correspondence analysis (CCA) and variance partition analyses (VPA) analytic methods to analyze effects of 35 soil properties, geographical and representative factors (listed in Supporting Information, Data S1) on bacterial communities across 16 representative cities in China, using 454-pyrosequencing technology. The objectives of this study were: (1) to determine dominant bacterial taxa in urban park soils; (2) to reveal the variation in the relative abundance and occurrence of dominant soil bacterial taxa throughout China; and (3) to determine the most important driving factors determining microbial community composition in urban park soils.

Materials and methods

Sampling sites

The 95 park soils were sampled from 16 provincial capital cities of China, including Harbin, Urumqi, Shenyang, Beijing, Xining, Jinan, Zhengzhou, Xi'an, Shanghai, Chengdu, Lhasa, Changsha, Fuzhou, Kunming, Guangzhou, and Haikou (Supporting Information, Fig. S1), with six samples per park in each city. As shown in Fig. S1, these 16 cities are distributed across China and consequently are located in different climatic zones, including the tropics, subtropics, temperate, frigid temperate and plateau regions. Furthermore, they have different levels of urbanization. The representative sampling site in each park was selected based on it having had no direct human activities for at least 10 years. Six samples were collected randomly within an undisturbed area of the park. Each of the samples from one park was composed of five subsamples which were collected from randomly selected locations of the top 10 cm of mineral soil within an area of about 100 m^2. Detailed information on sample collection is provided in Table 1.

After collection and compositing, all soil samples were stored in ice boxes before being posted or taken to Xiamen by air immediately. Upon arrival in Xiamen, the soil samples were divided into three. One subsample was stored at −80 °C for DNA extraction, one was stored moist at 4 °C for microbial biomass carbon (MBC) and

Table 1. Information for soil sampling parks from 16 cities in China

Cities	Parks	Latitude (°)	Longitude (°)	Plants
Harbin	Taiyangdao Park	45.79	126.60	Sample 1–sample 5: trees and grass sample 6: only trees
Urumqi	Renmin Park	43.89	87.56	Trees with few grass
Shenyang	Shenyang Arboretum	41.77	123.46	Trees with few grass
Beijing	Tiantan Park	39.88	116.41	Trees with few grass
Xining	Xining Renming Park	36.66	101.76	Big trees with grass
Jinan	Quancheng Park	36.65	117.02	Trees
Zhengzhou	Renming Park	34.76	113.66	Trees and grass
Xi'an	Botannical Garden	34.21	108.96	Trees with few grass
Shanghai	Luxun Park	31.28	121.48	Trees with few grass
Chengdu	Baihuatan Park	30.66	104.04	Trees with few grass
Lhasa	Norbulingka Park	29.66	91.09	Sample 1–sample 4: trees and grass sample 5, 6: tall grass
Changsha	Nanjiao Park	28.15	112.96	Trees with few grass
Fuzhou	West Lake Park	29.09	119.28	Trees with few grass
Kunming	Yuantongshan Park	25.06	102.71	Trees with few grass
Guangzhou	South China Botanical Garden	23.13	113.26	Trees and grass
Haikou	Haikou Park	20.04	110.34	Trees and grass

microbial biomass nitrogen (MBN) determination, and the remainder was air-dried and used for determination of physical and chemical properties (see below).

DNA extraction, PCR and pyrosequencing

Genomic DNA was extracted from 0.5 g soil from each sample using the Fast DNA SPIN kit for soil (Bio 101, MP) following the manufacturer's instructions with minor modifications (Peng *et al.*, 2010). The purity and the quantity of the DNA were determined by UV-Vis Spectrophotometer (ND-1000, NanoDrop) at 260 and 280 nm. All 260/280 ratios were found to be above 1.8. The DNA solution was stored at −20 °C until analyzed further.

The V4 and V5 regions of bacterial 16S rRNA genes were amplified on an ABI 9700 (Perkin-Elmer Applied Biosystems, Shenzhen, China) using the DNA extracted from the samples as template. The forward primer consisted of the 26-bp 454 adapter A, KEY (TCAG), a 10-bp index, followed by the 16-bp primer 515F (5′- CCATC TCATCCCTGCGTGTCTCCGAC-TCAG-barcode- GTGC CAGCMGCCGCGG- 3′). The reverse primer consisted of the 26-bp 454 adapter B, KEY (TCAG) and the 20-bp primer 907R (5′- CCTATCCCCTGTGTGCCTTGGCAGT C- TCAG- barcode-CCGTCAATTCMTTTRAGTTT- 3′). The targeted gene region has been reported to be the most appropriate for the accurate phylogenetic reconstruction of bacteria (Biddle *et al.*, 2008). PCR reactions were performed in a 50-μL volume containing 5 μL 10× PFX buffer, 2 μL $MgSO_4$, 2 μL dNTPs (10 mM each), 0.8 μL PFX polymerase, and 2 μL each of forward and reverse primers. Amplifications were run under the following cycling conditions: 3 min initial denaturation at 95 °C, followed by 30 cycles of denaturing at 95 °C for 15 s, annealing at 62 °C for 30 s, extension at 72 °C for 1 min, and completed with a final extension at 72 °C for 10 min. PCR products of DNA samples with different barcodes were mixed in equal concentration and pyrosequenced using the 454 GS FLX+ System.

Soil physical and chemical properties

Soil pH was determined with air-dried soil samples (sieved to < 2 mm) with a Dual Channel pH/Ion/Conductivity/Do Meter (X60, Fisher Scientific) at a soil : water ratio of 1 : 2.5 (w/v); total organic carbon (TOC) with a TOC analyzer (Shimadzu TOC-Vcph, Japan) with a solid sample module (SSM-5000A); and C, N and S with an elemental analyzer (Vario MAX CNS, Germany). Soil MBC and MBN were analyzed by the chloroform fumigation and extraction method as follows: two portions of moist soil (20 g) were weighed, the first (not fumigated) was immediately extracted with 80 mL of 0.5 M K_2SO_4 for 30 min by shaking, and filtered with quantitative filter paper; the second one was fumigated for 24 h at 25 °C with ethanol-free $CHCl_3$ and then extracted as described above. The filtrate after extraction was analyzed with a TOC analyzer which can determine C and N simultaneously (Shimadzu TOC-Vcph, Japan). Extractable major and trace metal concentrations were digested using a strong acid digestion method described by Lee *et al.* (2006), and determined with ICP-OES (Optima 7000DV; Perkin Elmer) and ICP-MS (Agilent 7500cx; Agilent Technologies, Inc., Tokyo, Japan).

Geographic and urbanization data

Geographical information, including latitude, annual average temperature (AT), annual average precipitation (AP), annual average relative humidity (AH) and city sunshine hours (CSH), is listed in Table S1. Urbanization factors including gross domestic product (GDP) per square kilometer (GDP km^{-2}), population density (PD), percentage of industry and service industry of GDP [$GDP_{(secondary\ +\ tertiary)}/GDP_{(total)}$], percentage of people working in industry and service industries [Work $force_{(secondary\ +\ tertiary)}$/work $force_{(total)}$] and the percentage of urbanized population (PUP) are listed in Table S2. Data on AT, AP, AH and CSH were averaged for the 4 years from 2007 to 2010 (China Statistical Yearbook, 2008, 2009, 2010, 2011). Data of GDP per square kilometer (GDP km^{-2}), PD (person km^{-2}), percentage of industry and service industries of GDP, percentage of people working for industry and service industries were collected and calculated from the China City Statistical Yearbook (2011). The PUP equals the population of people living in urban areas divided by the total population living in the metropolitan statistical area, with data collected from yearbooks of each province.

Processing pyrosequencing data

The data generated by 454-pyrosequencing was analyzed using the quantitative insights into microbial ecology (QIIME 1.6) toolkit (Caporaso *et al.*, 2010). After removing any low quality or ambiguous reads, qualified sequences were clustered into operational taxonomic units (OTUs) at 97% similarity level by default. The most abundant sequence from each OTU was selected as the representative sequence for that OTU and was assigned to taxonomy using an RDP classifier (version 2.2) (Wang *et al.*, 2007). To assess the internal (within-sample) complexity of individual microbial populations, the Shannon–Weaver index (H), PD whole tree and chao1 were calculated. Rarefaction curves were generated to compare the level of bacterial OTU diversity between samples. Phylogenetic trees were then built from all representative sequences using the FASTTREE algorithm (Price *et al.*, 2009). The differences in overall community composition between each pair of samples were determined using the unweighted UNIFRAC metric (Lozupone *et al.*, 2006), which calculates the distance between any pair of communities based on the fraction of unique branch length of their sequences in the tree (Lozupone & Knight, 2005).

Statistical analysis

To correct for survey effort (number of sequences analyzed per sample), we used a randomly selected subset of 2650 sequences per soil sample to compare relative differences between samples. Because only 1939, 2034, 1934 and 1334 sequences were generated for the soils Shenyang4, Shenyang5, Zhengzhou4 and Changsha4, respectively, these four samples were excluded when calculating the average for samples from each city.

To find out which factors are important in shaping bacterial community, CCA was conducted. Thirty-five factors were chosen to analyze their contribution to the variation of urban park soil microbial community by means of CCA. A variance test of significance and envfit function with 999 Monte Carlo permutations were used to remove environmental variables which did not contribute significantly to the total soil microbial community variance. Univariate statistical and correlation analyses were performed using SIGMAPLOT 12.0 (Systat Software, Inc., San Jose, CA). *P*-values < 0.05 were considered significant. Detrended correspondence analysis (DCA), CCA and VPA were performed using R (2.14.0, http://www.r-project.org/) with the community ecology package 'vegan (2.0-4)' (Oksanen, 2011).

Results

Geochemical characteristics

As shown in Fig. S2, soil samples from Urumqi had the highest total C, N and S, and Harbin the lowest. Total N contents in Urumqi, Kunming and Xi'an soil samples were a little higher than those in other cities. Kunming soil samples had the highest biomass C. Biomass N is more variable than biomass C within cities. Generally, Kunming, Urumqi and Xining had more biomass N than other cities.

Soil pH ranged from 3.8 to 8.2. Only 14 samples (six from Changsha, six from Guangzhou, one each from Fuzhou1 and Fuzhou2) had pH values below 6.0. Eight samples were neutral. More than 76% of the urban park soils were alkalescent or alkaline.

As shown in Table S1, cities in southern China generally had higher annual AP, higher annual average temperatures, and fewer sunshine hours compared with cities in the north. Cities in the east had higher average annual precipitation compared with the west. Although located in south China, Lhasa had the highest number of sunshine hours, lower AT and lower annual AP due to its high elevation.

All the urban population percentages, except that of Lhasa, were higher than 60%. Shanghai, Beijing and Guangzhou are the three most urbanized cities, with

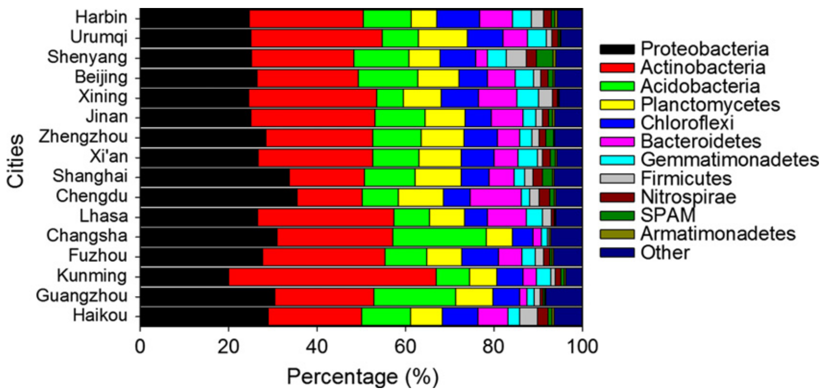

Fig. 1. Average relative abundances of bacterial community composition at phylum level detected in urban park soils in 16 cities in China. Abundance is presented in terms of an average percentage of the total effective bacterial sequences of six samples from each city, classified by RDP Classifier at a confidence threshold of 97%. 'Other' refers to the taxa with a maximum abundance of < 1% in any sample.

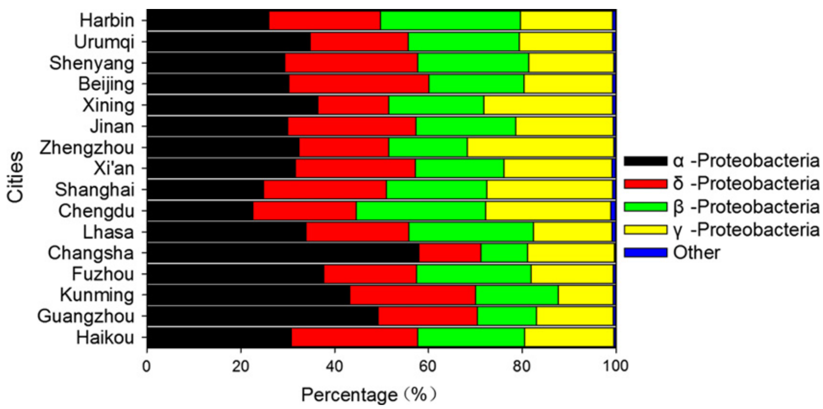

Fig. 2. Average relative abundances of *Proteobacteria* composition by class in each city. The abundance is presented in terms of an average percentage of the total number of *Proteobacteria* sequences of six samples of each city, classified at a confidence threshold of 97%. 'Other' refers to the taxa with a maximum abundance of < 1% in any sample.

urban population percentages of 89.30%, 86.00% and 84.13%, respectively.

Composition of bacterial communities– dominant taxa in urban park soils

A total of 980 019 sequences were generated from 95 samples (one sample from Kunming had a failed PCR). After filtering, 574 442 high quality archaeal and bacterial 16S rRNA gene sequences, about 59% of the total sequences, remained (Table S3), and for each sample 1334–19 043 effective sequence tags were obtained. More than 99.5% of the sequences were from bacteria. High quality sequences were clustered into OTUs at 97% similarity by default. After assignment, 48 824 OTUs were recovered from all the samples; individual samples contained 509–4753 OTUs. Individual samples only contained between 1.0% and 9.7% of the total OTUs, which would seem to indicate that the samples are quite different from each other.

Rarefaction was performed to show the α-diversity of samples at a sampling depth of 2650 sequences (Fig. S3). The results of Shannon–Weaver index (H), PD whole tree and observed species all showed that samples from Changsha had the lowest bacterial diversity, followed by samples from Guangzhou; Chengdu samples had the highest α-diversity (see Table S4). The α-diversities of most urban park soil samples were not significantly different. Correlation analysis showed that all the α-diversity indexes were significantly correlated with pH, As, Sb, Mn, Tl and Mg, but not with any other factor (see Table S5).

As shown in Fig. 1, at phylum level, the structures of the microbial communities differed in terms of both the predominant phylum and the relative abundance of each phylum. *Proteobacteria, Actinobacteria, Acidobacteria, Planctomycetes, Chloroflexi* and *Bacteroidetes* were the six most abundant phyla. In all samples, *Proteobacteria* and *Actinobacteria* were the two most abundant phyla, together accounting for 48–67% of all bacterial sequences obtained in each city in the survey. In particular, in four city soil samples (Urumqi, Xining, Lhasa, Kunming) the predominant phylum was *Actinobacteria*, with Kunming having the highest abundance of this phylum (46.99%). *Proteobacteria* constituted the highest percentage of the phyla found in seven other city soil samples (Beijing, Zhengzhou, Shanghai, Chengdu, Changsha, Guangzhou, Haikou). In contrast, the abundance of *Proteobacteria* and *Actinobacteria* in the remaining five city samples (Harbin, Jinan, Shenyang, Xi'an

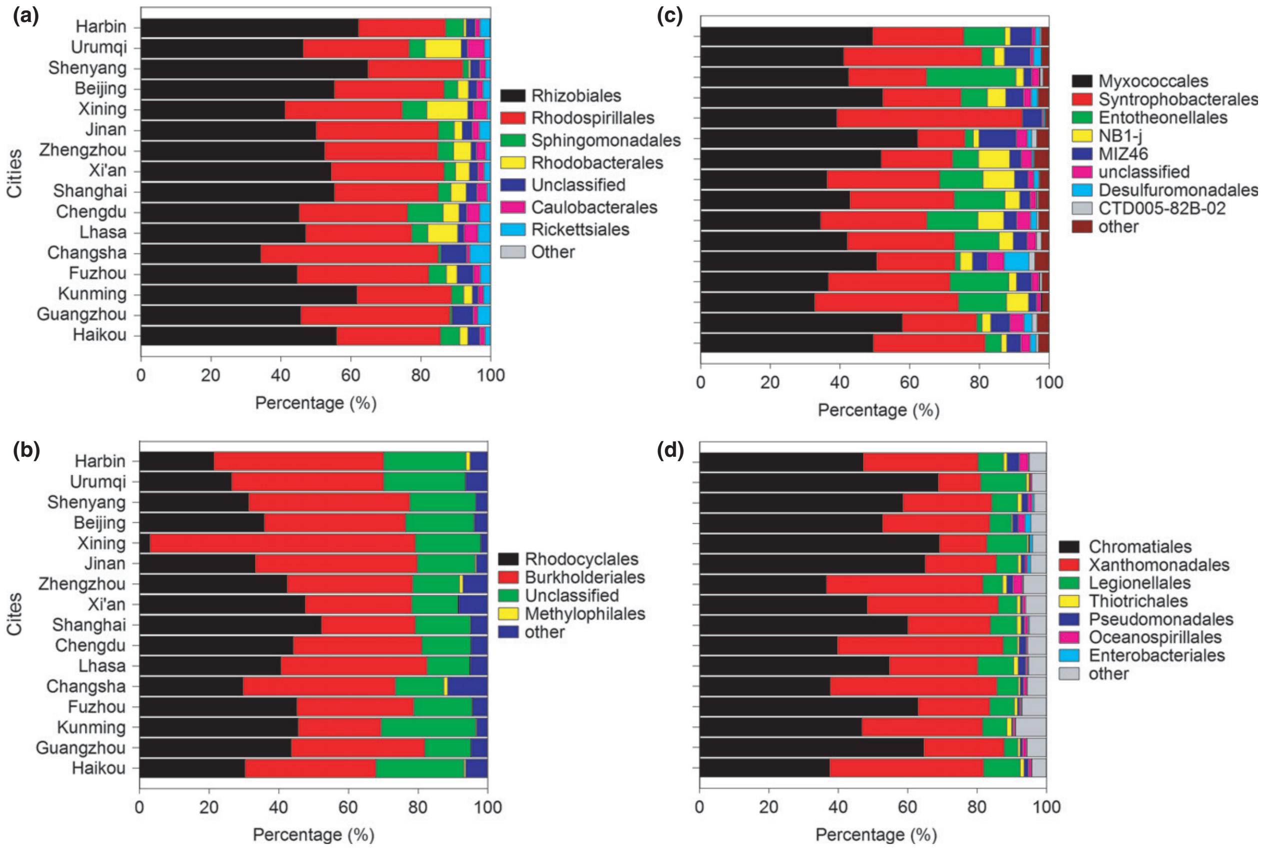

Fig. 3. *Alphaproteobacteria* (a), *Betaproteobacteria* (b), *Gammaproteobacteria* (c) and *Deltaproteobacteria* (d) composition by order in each city. Abundance is presented in terms of percentage of *Alphaproteobacteria, Betaproteobacteria*, *Gammaproteobacteria* and *Deltaproteobacteria* sequences detected relative to total sequences in soils from parks in each city. 'Other' refers to the taxa with a maximum abundance of < 1% in any sample.

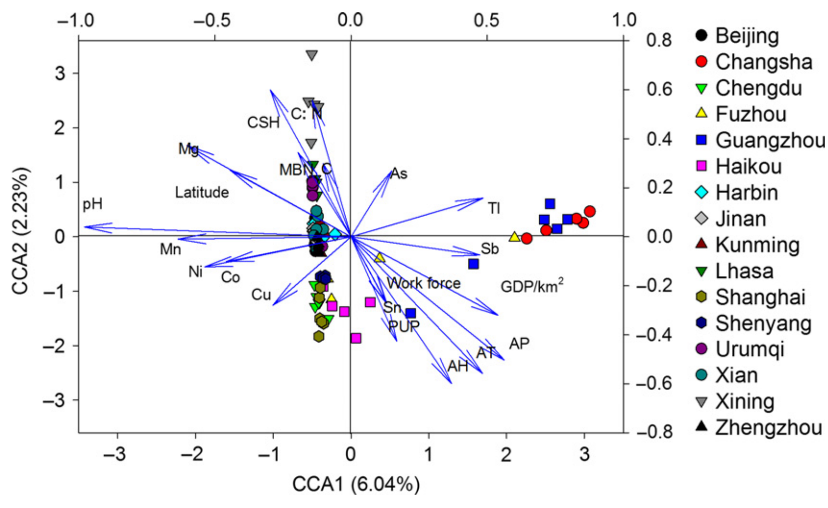

Fig. 4. CCA of pyrosequencing data and environmental factors. Arrows indicate the direction and magnitude of environmental factors associated with bacterial community structure in different samples. The result was processed with OTU metrics and environmental factors. PUP, is percentage of urbanized population; AP, annual average precipitation; AT, annual average temperature; AH, annual average relative humidity; CSH, city sunshine hours.

and Fuzhou) was low. As the third most abundant phylum, the *Acidobacteria* abundances varied from 6.01% (in Xining) to 21.12% (in Changsha).

Further analysis of *Proteobacteria* was conducted to determine the differences at class levels (Fig. 2). *Proteobacteria* was mainly composed of four subdivisions: *Alpha-*

proteobacteria, Betaproteobacteria, Gammaproteobacteria, and Alphaproteobacteria. Alphaproteobacteria, which was the predominant class in all samples, was most abundant in Changsha soil.

Among Alphaproteobacteria, Rhizobiales and Rhodospirillales were found to be two of the most abundant orders in all samples (Fig. 3). Notably, in Changsha urban park soil samples, Rhodospirillales abundance was higher than Rhizobiales. Rhodocyclales and Burkholderiales were dominant in Betaproteobacteria. In Xining soils, the abundance of Rhodocyclales was abnormally low, whereas Burkholderiales was markedly higher than in other cities. Myxococcales and Syntrophobacterales were the most abundant in Deltaproteobacteria, whereas

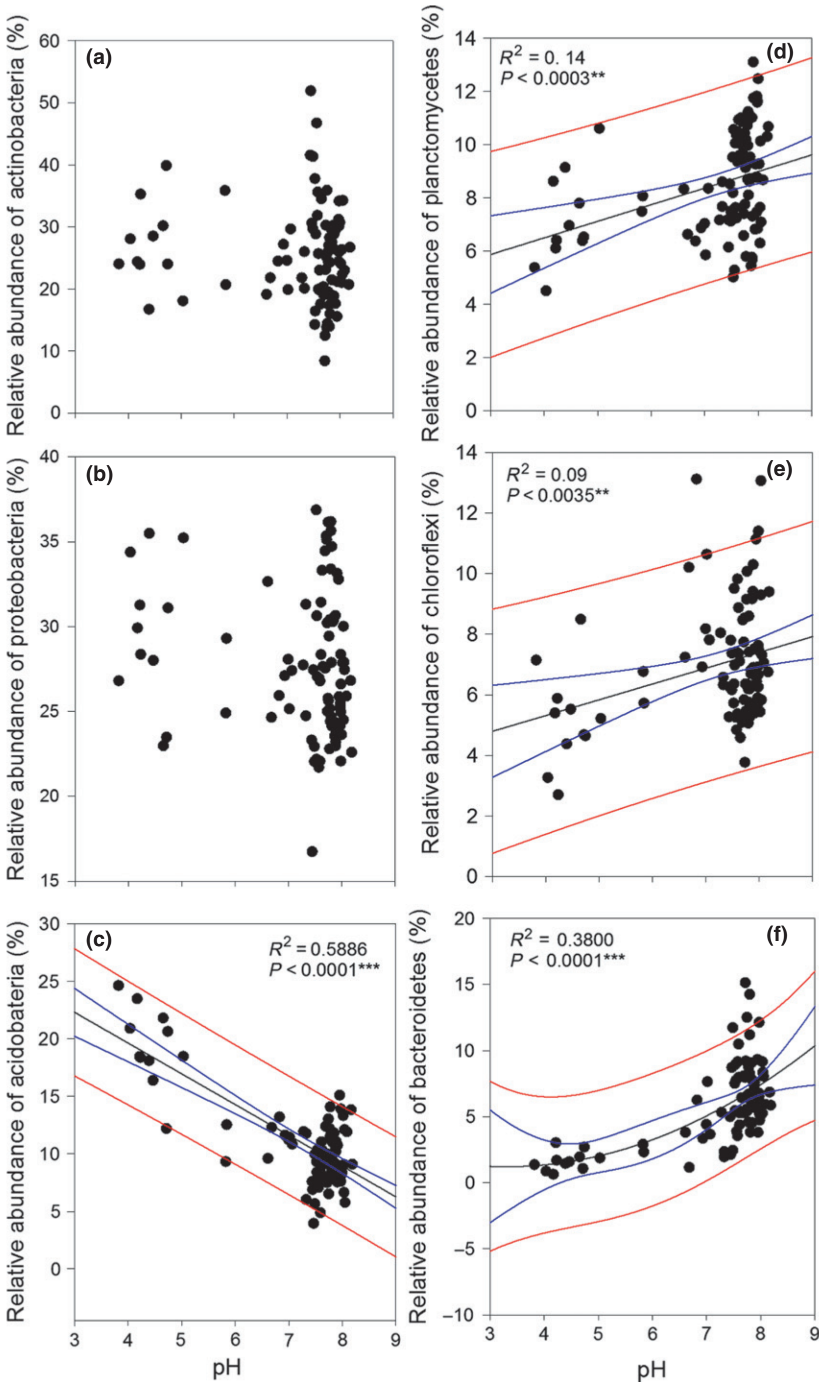

Fig. 5. Relationships between relative abundances of six dominant bacterial phyla and soil pH. Linear or quadratic regressions were used to test the relationship between the taxa relative abundances and soil pH. Adjusted R^2 values with the associated P-values are shown for each taxonomic group. Black lines represent the fit model to the data, blues ones represent 95% confidence bands and red ones represent 95% prediction bands. Relationship between pH and (a) *Actinobacteria*, (b) *Proteobacteria*, (c) *Acidobacteria*, (d) *Planctomycetes*, (e) *Chloroflexi* and (f) *Bacteroidetes*.

Chromatiales and *Xanthomonadales* dominated in *Gammaproteobacteria*.

Using the RDP Classifier, although a total of 275 orders were obtained (see Table S6), only 29 orders were common to all samples (see taxa tagged in bold in Table S6). These 29 orders accounted for 64% (Shenyang6) to 90% (Fuzhou1) of classified sequences in each sample, which indicated that the main taxa in the samples from each city were similar.

Variations among samples

DCA was performed to examine the overall variation among bacterial communities of the 91 samples analyzed. Figure S4 shows that samples were scattered in two main sections along the first axis. The right section included samples from Guangzhou, Changsha and some from Fuzhou and Haikou; the remaining samples were clustered closely in the left section. Along the second axis there was no clear boundary among all the samples; however, samples with similar latitude and climate factors were located close together (e.g. Shanghai and Chengdu; Beijing, Jinan and Xining).

Correlations between environmental parameters and bacterial community compositions – driving factors of variation

CCA was performed to reveal possible relationships between microbial community composition and environmental parameters (Fig. 4). Based on the variance test of significance and envfit function with 999 Monte Carlo permutations, 21 significant environmental variables, including TC, C : N ratio, MBN, pH, Cu, As, Ni, Sb, Sn, Co, Mn, Tl, Mg, AP, AT, AH, CSH, latitude, GDP km^{-2}, Work $force_{(secondary\ +\ tertiary)}$/work $force_{(total)}$ and PUP, were selected in the CCA biplot. Like the results of DCA, along the first axis of CCA plot, there are two separate main sections, mainly determined by pH, Mn, Ni, Sb and Tl. Xining, Urumqi and Lhasa clustered closely on the second axis, whereas the rest of the samples scattered closely with Haikou samples at the far edge. The C : N ratio and PUP contributed more to the distribution along the second axis. The rest of the factors were distributed equally in both axes.

CCA results showed that pH had the strongest effects on bacterial community composition (Fig. 4). Geographical factors contributed significantly to the variance. In contrast, urbanization factors did not have significant effect on bacterial community composition compared with pH, AP, AT, AH, CSH and latitude.

To test and verify the effects of pH on bacterial community composition, a correlation analysis between pH and the relative abundance of six abundant phyla was done (Fig. 5). The results showed that the relative abundances of *Planctomycetes*, *Chloroflexi* and *Bacteroidetes* were positively correlated, with pH and *Acidobacteria* abundance having a significant negative correlation with pH. However, what surprised us is that as the two most abundant phyla, the abundance of neither *Proteobacteria* or *Actinobacteria* had any obvious correlation with pH, and these phyla are negatively correlated with each other (Fig. S5).

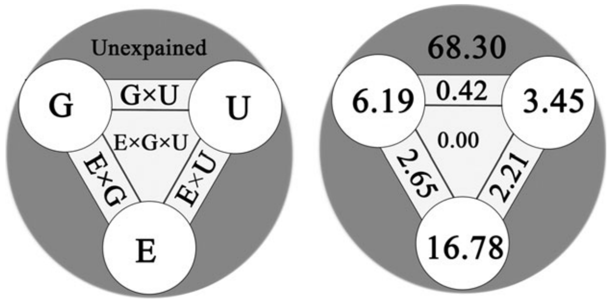

Fig. 6. Variance partitioning analysis of microbial community explained by edaphic properties (E), geographic locations (G) and urbanization (U). The apexes of the triangles represent the variation explained by each factor alone. The sides of the triangles represent interactions of any two factors, and the middle of the triangles represents interactions of all three factors. The gray part of the biggest circle stands for the unexplained part. The data are percentages of each part of the total bacterial community composition variance (%).

The contributions of edaphic properties, urbanization factors and geographic locations to microbial community variance were assessed by variance partitioning analyses. Figure 6 showed that 31.70% of the variance could be explained by these three groups of factors. Soil properties, geographic location and urbanization indexes independently explained 16.78%, 6.19% and 3.45% of the total bacterial community variance, respectively. Interaction between edaphic properties and geographic locations explained 2.65% of the variance, and interaction between edaphic properties and urbanization indexes, 2.21%. Interactions among the three components had much less influence on bacterial community composition and therefore can be neglected.

Discussion

In this study, *Proteobacteria, Actinobacteria, Acidobacteria, Planctomycetes, Chloroflexi* and *Bacteroidetes* were the six most abundant phyla in urban park soils in China. The observed species within samples from each city did not show significant differences, except for those from Changsha and Guangzhou. Twenty-nine shared orders account for as high as 64–90% of total classified sequences in all soil samples, which means the main taxa in the communities are constant despite the very different relative abundances between soils and the fact that each soil has its own rare taxa. These results indicate that the bacterial community composition variance across continental scales in China is mainly due to the relative abundances of each taxa instead of differences in taxa types. The predominant phylum in soils from Urumqi, Xining, Lhasa and Kunming was *Actinobacteria*. Coincidently, these four cities are all in western China with elevations above 680 m, and have more MBN than other cities. It is suggested that elevation, MBN and temperature may be important factors affecting the relative abundances of *Actinobacteria* and *Proteobacteria*.

To the best of our knowledge, this is the first research conducted to study the effect of urbanization factors on soil microbes. This study reveals that although α-diversity was not significantly correlated with geographical factors and urbanization indexes, CCA and VPA results show that both these factors are important in shaping urban park soil bacterial communities in China. As many previous studies have shown, soil properties are also important driving factors of bacterial communities of urban park soils in this study, and pH is a primary factor shaping soil bacterial community composition (Lauber *et al.*, 2009). In general, the correlation results between the abundances of the six main phyla and pH indicated that pH affects soil bacterial community composition by increasing or decreasing the abundances of different taxa to different extents. However, it is surprising that pH had no obvious correlation with the two most abundant phyla, *Proteobacteria* and *Actinobacteria*, which were negatively correlated with each other. It can be speculated that *Proteobacteria* and *Actinobacteria* are less sensitive to pH, and thus are dominant in both acidic and alkaline soils and are negatively correlated due to their higher relative abundances compared with the other phyla. This means that in order to predict the likely composition of soil bacterial communities effectively, additional factors that could influence the abundance of *Proteobacteria* and *Actinobacteria* should be considered. Few studies have estimated how much of the bacterial community composition could be explained by pH at a continental scale. Edaphic properties, including pH, could only explain 16.78% of the variance in our investigation, suggesting that although pH is an important factor in shaping soil bacterial communities, it cannot be concluded that urban park soil bacterial community composition is predictable by pH alone, at least at a continental scale.

It is well known that plants and animals exhibit a regular geographical pattern across latitude. Unlike the results of Fierer & Jackson (2006), who showed that there was no apparent regular latitudinal gradient in bacterial diversity, CCA and VPA results in this study showed that geographical locations play a part in predicting bacterial community composition. This indicates that, similar to animals and plants, microbes may also exhibit some geographical distribution patterns along latitude or elevation, but the pattern may be much more easily disturbed by soil properties and other environmental factors. Few studies have been carried out on the effect of meteorological factors on soil microbes. Annual AP, annual average temperature, annual AH and CSH are determined by geographical location, and these factors thus were categorized as geographical factors in this study. We found that these factors contributed to soil microbial community variance.

Urbanization factors are shown to be important in predicting bacterial community composition in this study. As stated above, these urban park soils were chosen to avoid direct disturbances from human activities. However, our results indicated that urbanization processes can still exert some influences on soil bacterial community composition even on urban soil which has not been directly polluted or artificially changed.

Based on VPA results, 31.70% of the community variance could be explained by these three groups of factors including soil property factors, geographical factors and urbanization factors. It is reasonable to expect that some additional factors, such as water-retention capacity, soil temperature, soil texture, soil cover and other chemicals, play significant roles in mediating bacterial community structures in urban park soil.

In conclusion, the present study shows that the dominant bacterial taxa in urban park soils in China are *Proteobacteria, Actinobacteria, Acidobacteria, Planctomycetes, Chloroflexi* and *Bacteroidetes*. The bacterial diversity and community composition are driven by many factors. Soil properties are the most important driving factors in shaping urban park soil bacterial community composition, and geographic locations and urbanization indexes are also significant predicting factors. Nearly 70% of the total variance could not be explained by the three groups of tested factors, suggesting that microbial communities in urban park soils are highly variable and cannot be easily predicted by common factors.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Data S1. Material and methods.
Fig. S1. Map of sampling cities in China.
Fig. S2. Chemical and biological properties of soil samples collected from 16 urban parks in China.
Fig. S3. α-Diversity comparison between cities.
Fig. S4. Detrended correspondence analysis (DCA) of pyrosequencing data.
Fig. S5. Relationship between relative abundances of *Proteobacteria* and *Actinobacteria*.
Table S1. Geographical information for sampling sites of 16 urban parks from 16 cities of China.
Table S2. Urbanization indexes of 16 sampling cities in China.
Table S3. Overview of the results of quality filtering, chimera detection and analysis for all 95 soil samples collected from 16 city parks of China.
Table S4. α-Diversity at the depth of 2650 sequences of urban park soil bacterial from 16 cities of China.
Table S5. Correlation between soil bacterial α-diversity and soil properties of soil samples collected from 16 city parks of China.
Table S6. Classified orders from all the soil samples collected from 16 city parks of China.